

Информация

Докладчик

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Цель работы

Целью данной лабораторной работы является ознакомление с основами набора математических выражений в LaTeX.

The purpose of this lab work is to learn how to typeset mathematical formulas and equations using LaTeX math mode and related packages.

Задание

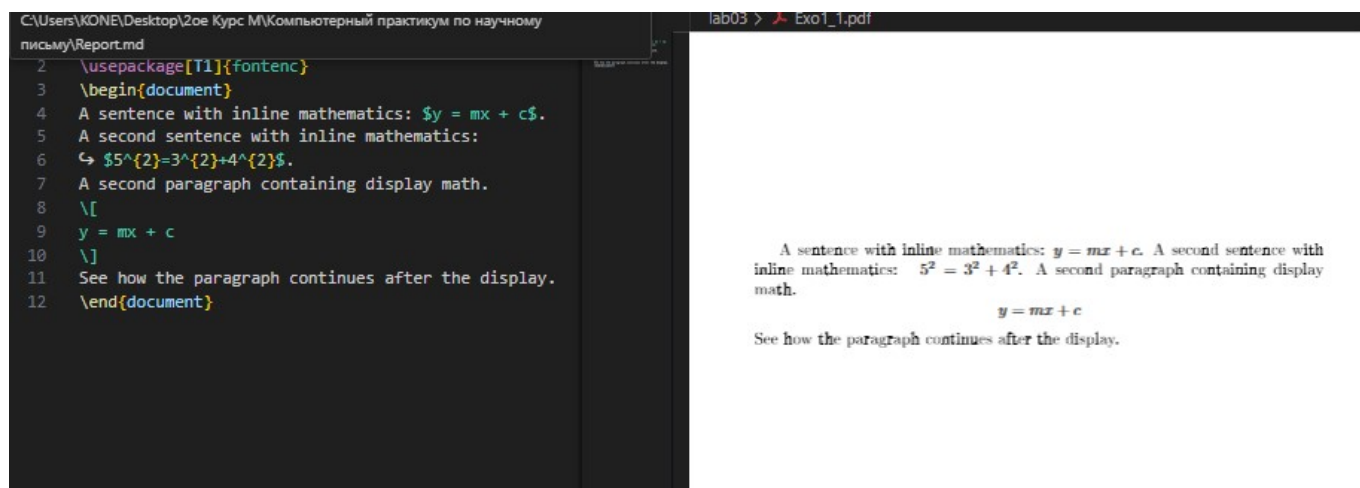
1. Study inline and display math modes.
2. Use the `amsmath` package to align and format equations.
3. Apply different math fonts.
4. Use `mathtools` for advanced formatting.
5. Try bold math and Unicode math.
6. Perform the exercises with examples.

Теоретическое введение

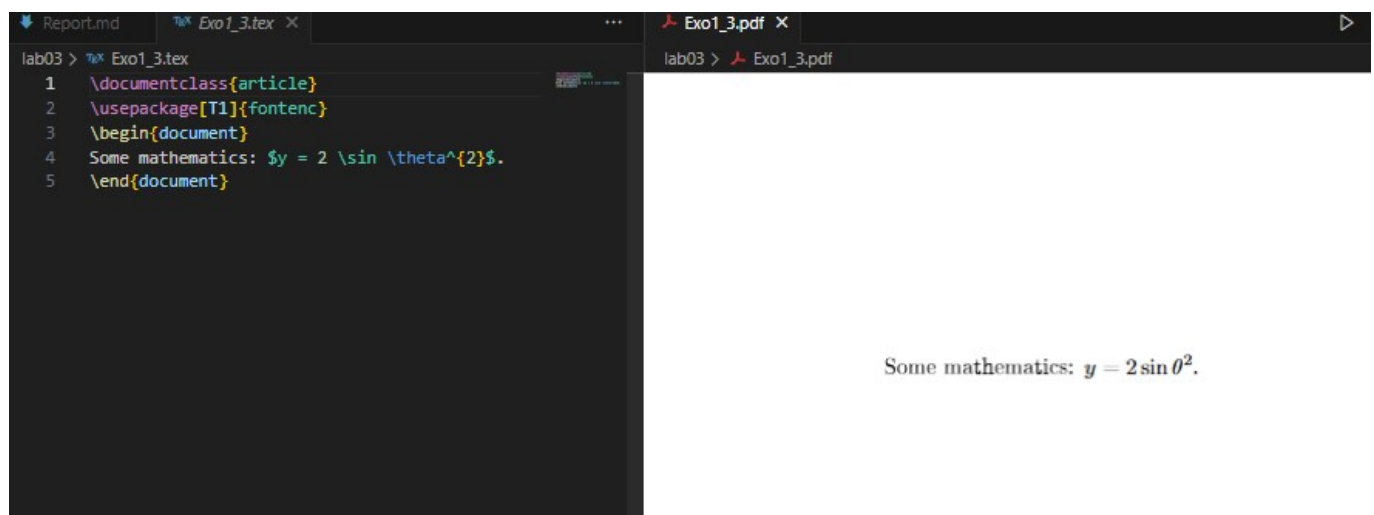
3.1 Математический режим / Math mode

В LaTeX существует два математических режима: **inline** и **display**.

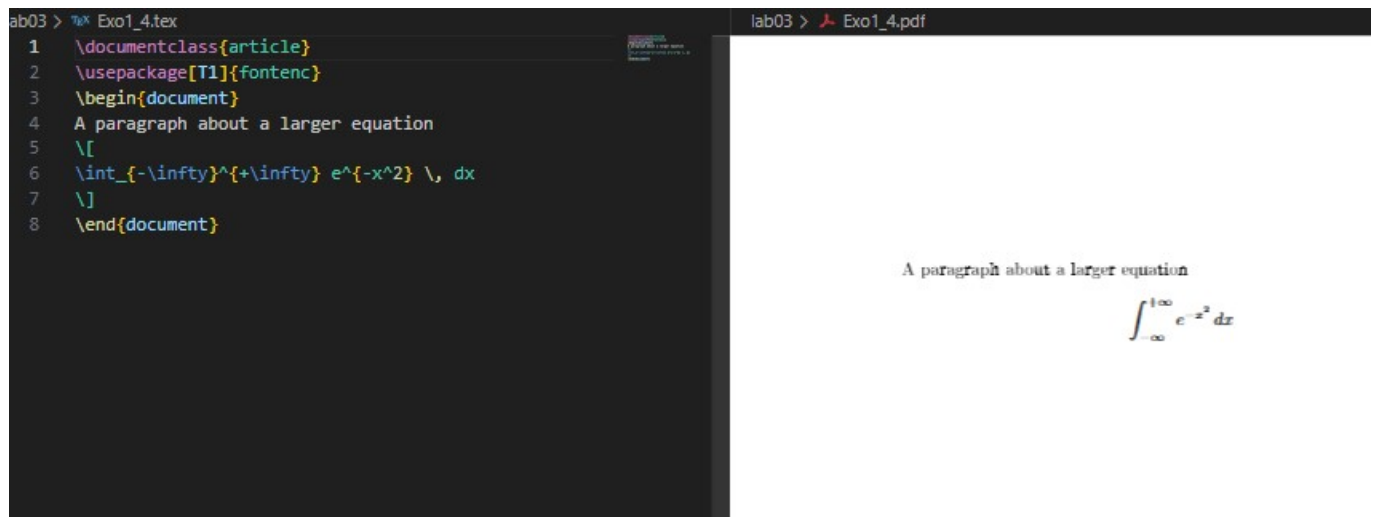
In LaTeX there are two main math modes: inline (within text) and display (centered block).

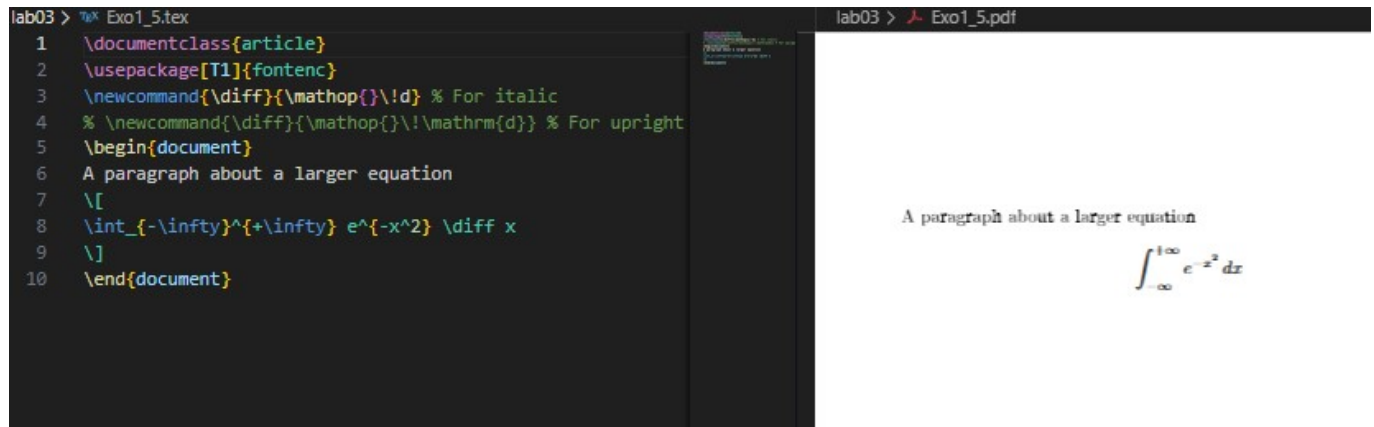


3.1.1 Inline math mode and mathematical notation



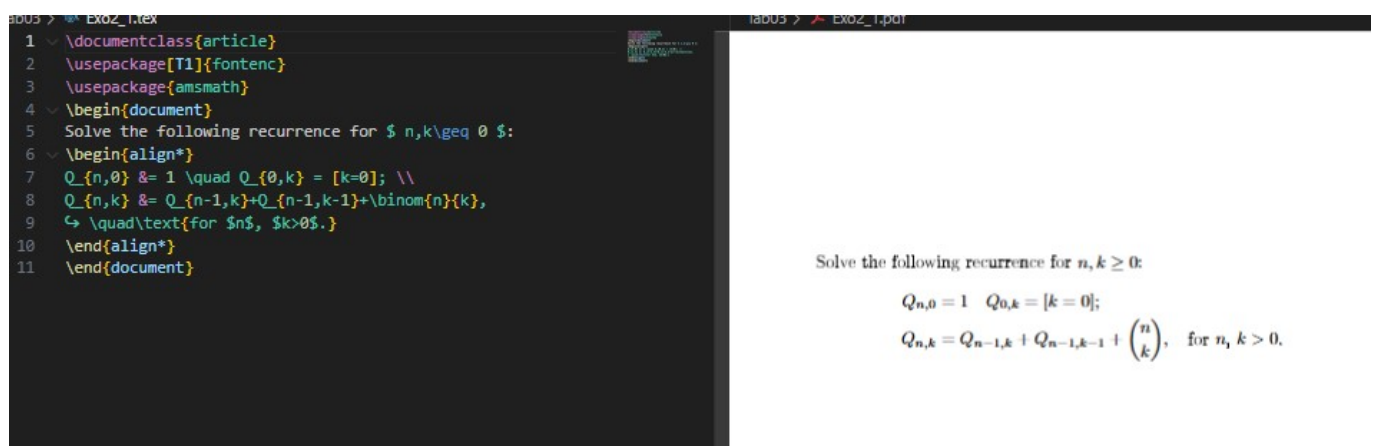
3.1.2 Display mathematics

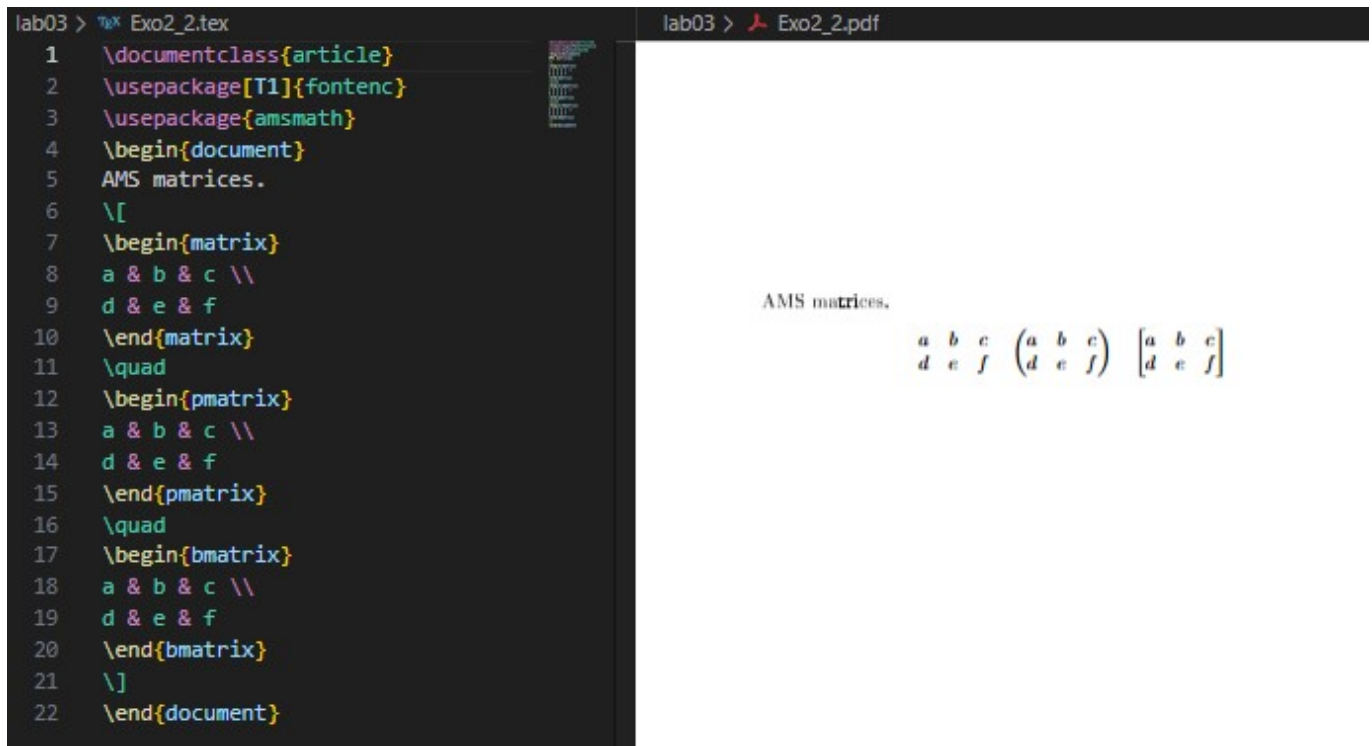




3.2 Пакет amsmath / The amsmath package

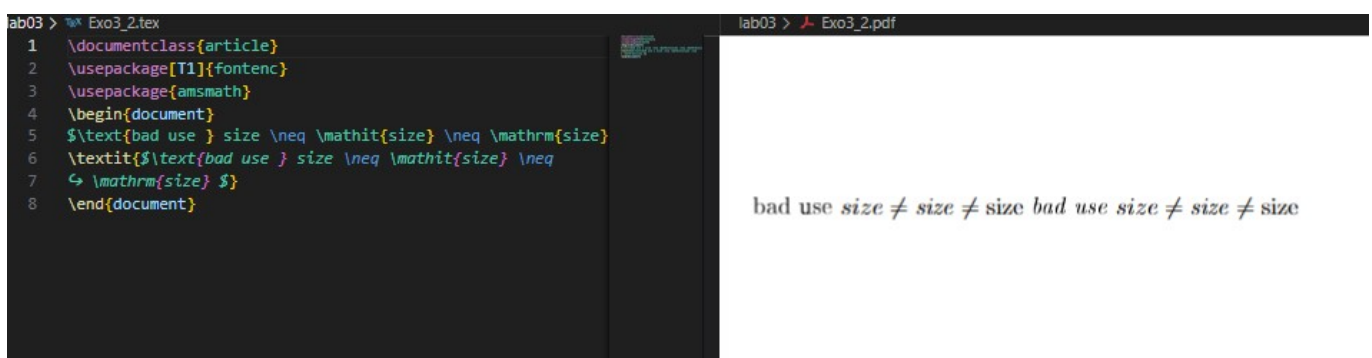
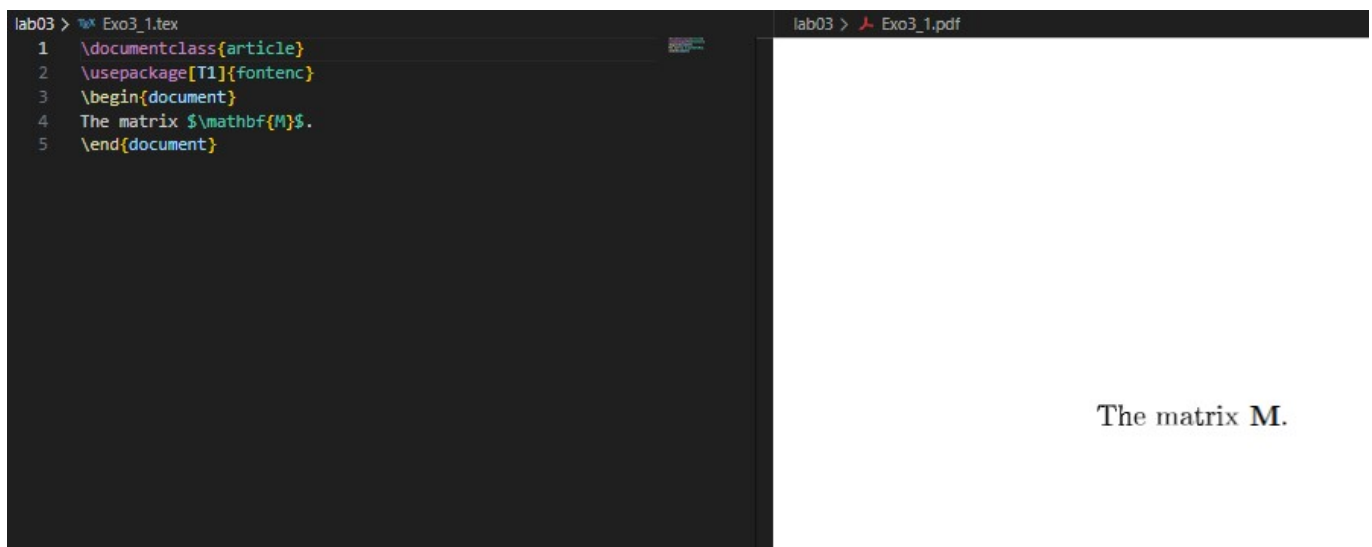
Пакет **amsmath** расширяет стандартные возможности для набора формул и выравнивания уравнений. The **amsmath** package enhances math typesetting and alignment.





3.3 Шрифты в математическом режиме / Fonts in math mode

В математике разные шрифты обозначают разные типы объектов. Different font commands give different styles and meanings.



3.4 Дополнительные выравнивания / Further amsmath alignments

Environments like `gather` and `multline` are used for multi-line equations.

```

lab03 > %\ Exo4_1.tex
1 \documentclass[a4paper]{article}
2 \usepackage[T1]{fontenc}
3 \usepackage{amsmath}
4 \begin{document}
5 \begin{gather}
6 P(x)=ax^5+bx^4+cx^3+dx^2+ex+f\\
7 x^2+x=10
8 \end{gather}
9 Multline
10 \begin{multline*}
11 (a+b+c+d)x^5+(b+c+d+e)x^4 \\
12 +(c+d+e+f)x^3+(d+e+f+a)x^2+(e+f+a+b)x\\
13 + (f+a+b+c)
14 \end{multline*}
15 \end{document}

```

```

lab03 > \ Exo4_1.pdf

```

$$P(x) = ax^5 + bx^4 + cx^3 + dx^2 + ex + f \quad (1)$$

$$x^2 + x = 10 \quad (2)$$

Multline

$$(a + b + c + d)x^5 + (b + c + d + e)x^4 + (c + d + e + f)x^3 + (d + e + f + a)x^2 + (e + f + a + b)x + (f + a + b + c)$$

3.4.1 Columns in math alignments

```

lab03 > %\ Exo4_2.tex
1 \documentclass{article}
2 \usepackage[T1]{fontenc}
3 \usepackage{amsmath}
4 \begin{document}
5 Aligned equations
6 \begin{align*}
7 a &= b+1 & c &= d+2 & e &= f+3 \\
8 r &= s^2 & t &= u^3 & v &= w^4
9 \end{align*}
10 \end{document}

```

```

lab03 > \ Exo4_2.pdf

```

Aligned equations

$a = b + 1$	$c = d + 2$	$e = f + 3$
$r = s^2$	$t = u^3$	$v = w^4$

```

lab03 > %\ Exo4_3.tex
1 \documentclass{article}
2 \usepackage[T1]{fontenc}
3 \usepackage{amsmath}
4 \begin{document}
5 \begin{itemize}
6 \item
7 \begin{aligned}[t]
8 a&=b\\
9 c&=d
10 \end{aligned}
11 \item
12 \begin{aligned}
13 a&=b\\
14 c&=d
15 \end{aligned}
16 \end{itemize}
17 \end{document}

```

```

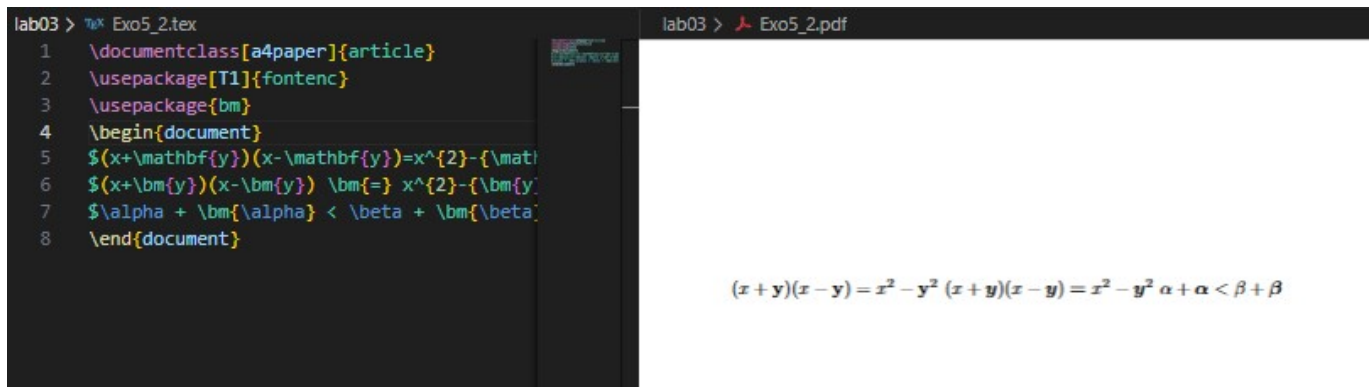
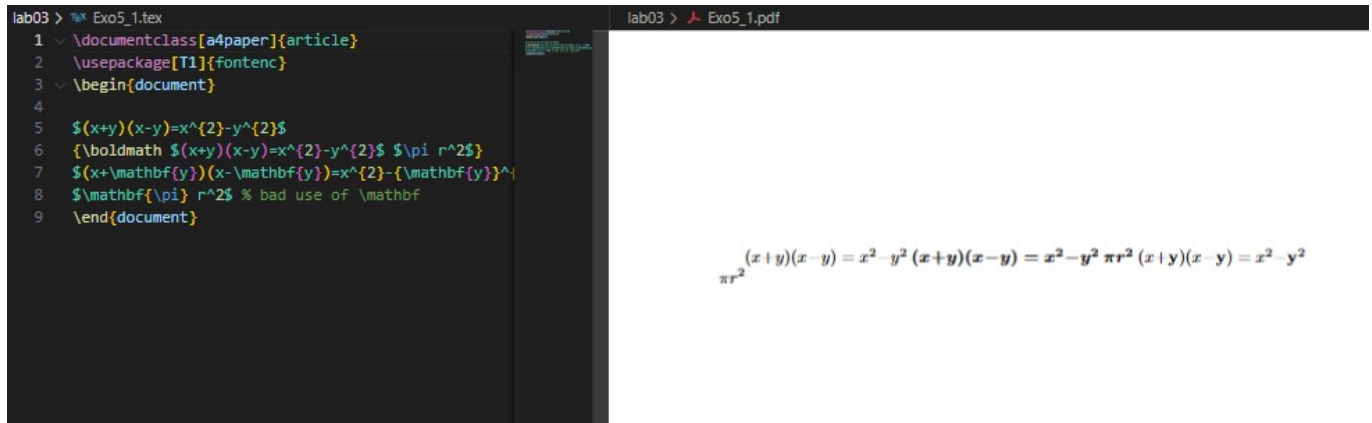
lab03 > \ Exo4_3.pdf

```

- $a = b$
- $c = d$
- $a = b$
- $c = d$

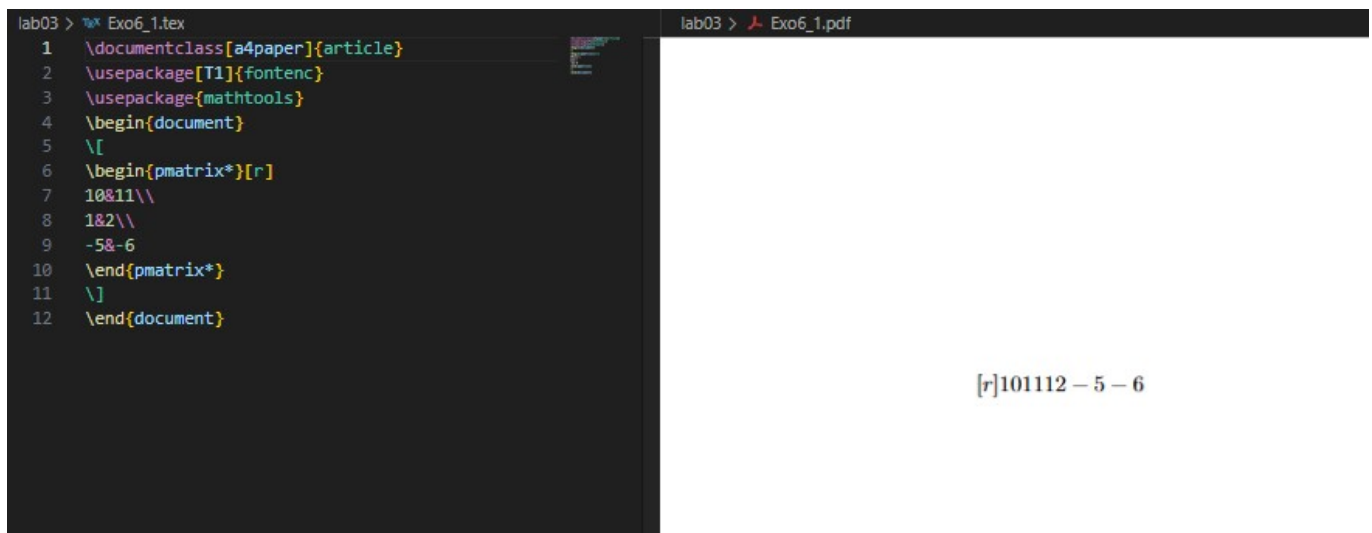
3.5 Жирный шрифт в формулах / Bold Math

To bold entire or partial equations, we can use `\boldmath` or the `bm` package.



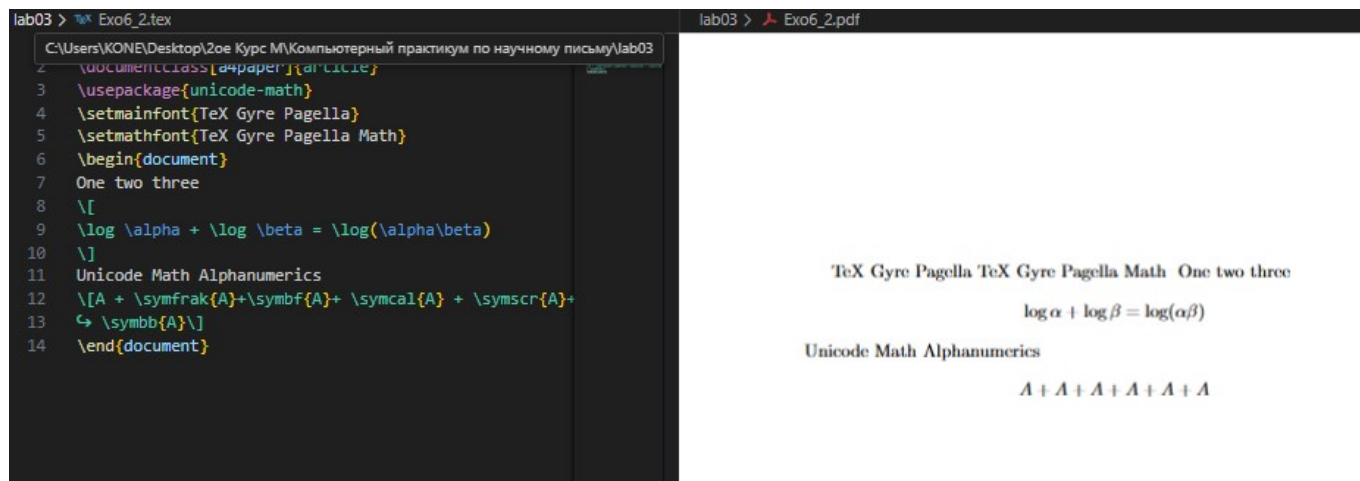
3.6 Пакет Mathtools / Mathtools package

`mathtools` builds upon `amsmath` and provides extended features like column alignment in matrices.



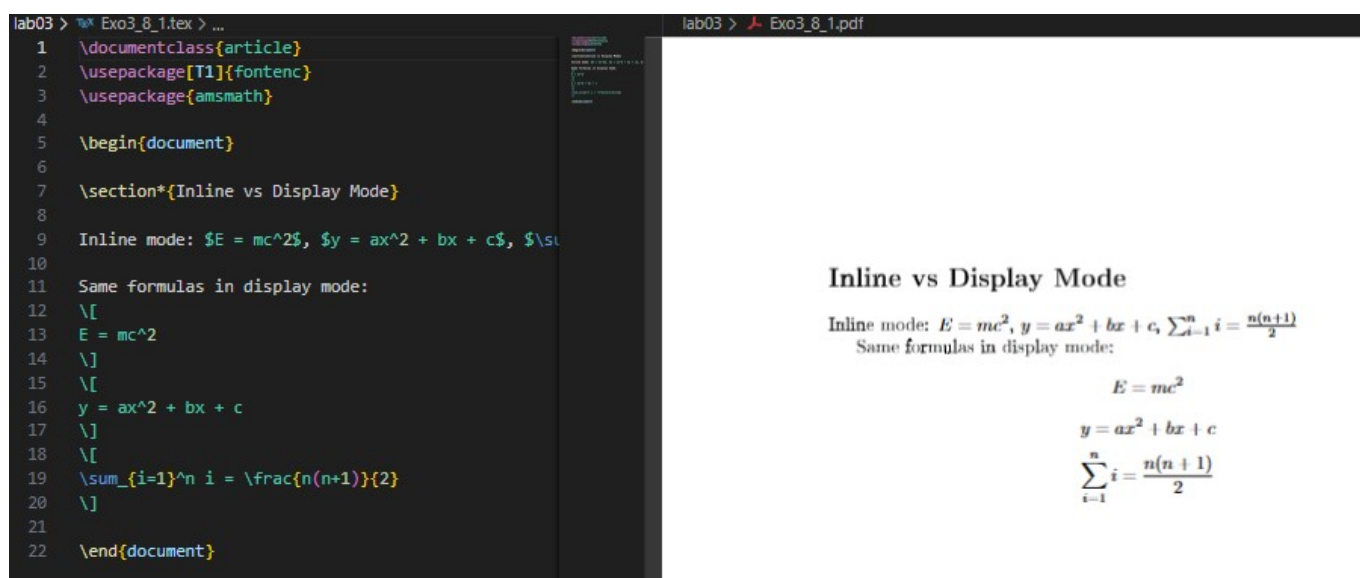
3.7 Юникодная математика / Unicode Math

Using `unicode-math` with OpenType fonts allows modern mathematical typesetting.

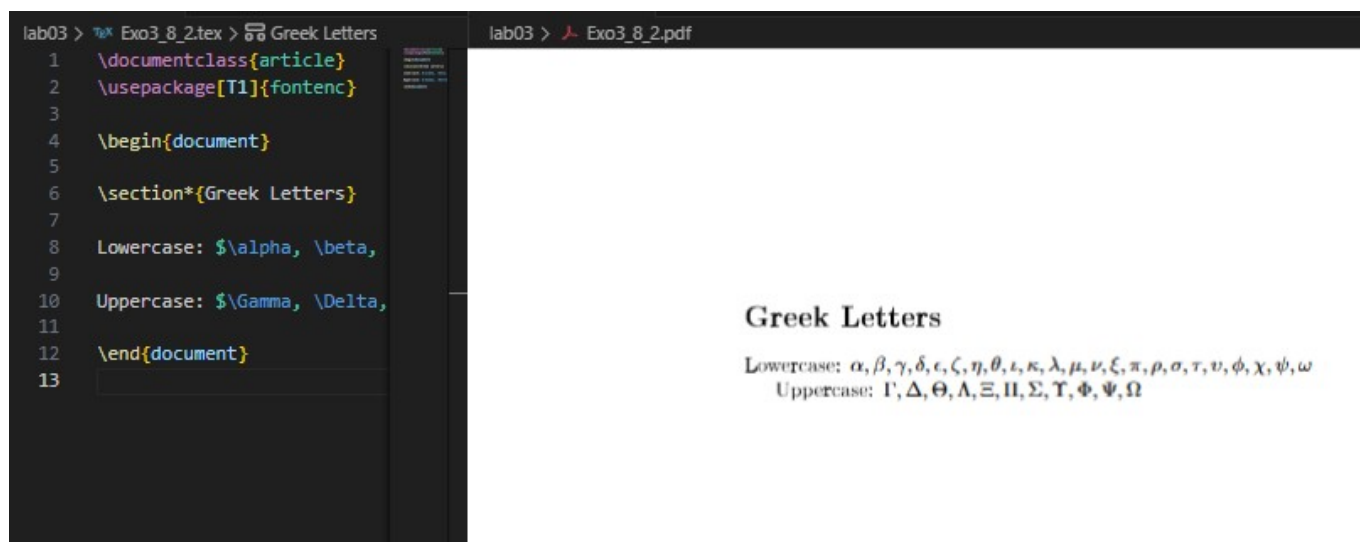


Выполнение лабораторной работы

1. Переключение между режимами / Switching between math modes



2. Греческие буквы / Greek letters



3. Комбинирование шрифтов / Combining fonts

```

lab03 > \TeX Exo3_8_3.tex > Font Commands in Math Mode
1 \documentclass{article}
2 \usepackage[T1]{fontenc}
3 \usepackage{amsmath}
4
5 \begin{document}
6
7 \section*{Font Commands in Math Mode}
8
9 Normal font: $abcABC123$
10
11 Roman: $\mathrm{abcABC123}$
12
13 Text italic: $\mathit{abcABC123}$
14
15 Bold: $\mathbf{abcABC123}$
16
17 Sans serif: $\mathsf{abcABC123}$
18
19 Typewriter: $\mathtt{abcABC123}$
20
21 \section*{Nesting Commands}
22
23 Nesting test: $\mathbf{\mathit{text}}$
24
25 Combination: $\mathbf{x} + \mathit{y} = \mathsf{z}$
26
27 \end{document}
28

```

lab03 > Exo3_8_3.pdf

Font Commands in Math Mode

Normal font: $abcABC123$
 Roman: $\mathrm{abcABC123}$
 Text italic: $\mathit{abcABC123}$
 Bold: $\mathbf{abcABC123}$
 Sans serif: $\mathsf{abcABC123}$
 Typewriter: $\mathtt{abcABC123}$

Nesting Commands

Nesting test: $\mathbf{\mathit{text}}$
 Combination: $\mathbf{x} + \mathit{y} = \mathsf{z}$

4. Параметры класса документа для уравнений / Equation alignment

```

lab03 > \TeX Exo3_8_4.tex > Left-Aligned Equations
1 \documentclass[fleqn]{article}
2 \usepackage[T1]{fontenc}
3 \usepackage{amsmath}
4
5 \begin{document}
6
7 \section*{Left-Aligned Equations}
8
9 Simple equation:
10 \[
11 E = mc^2
12 \]
13
14 Complex equation:
15 \[
16 \int_{-\infty}^{\infty} e^{-x^2} \, dx = \sqrt{\pi}
17 \]
18
19 Numbered equation:
20 \begin{equation}
21 a^2 + b^2 = c^2
22 \end{equation}
23
24 \end{document}
25

```

lab03 > Exo3_8_4.pdf

Left-Aligned Equations

Simple equation:

$$E = mc^2$$

Complex equation:

$$\int_{-\infty}^{\infty} e^{-x^2} \, dx = \sqrt{\pi}$$

Numbered equation:

$$a^2 + b^2 = c^2$$

```

lab C:\Users\KONE\Desktop\2oe Курс М\Компьютерный практикум по научному письму\lab03
\Exo3_8_5.tex (preview @)
2 \usepackage[T1]{fontenc}
3 \usepackage{amsmath}
4
5 \begin{document}
6
7 \section*{Left-Side Equation Numbers}
8
9 Left-numbered equation:
10 \begin{equation}
11 e^{i\pi} + 1 = 0
12 \end{equation}
13
14 Another equation:
15 \begin{equation}
16 F = G\frac{m_1 m_2}{r^2}
17 \end{equation}
18
19 \end{document}
20
21

```

lab03 > Exo3_8_5.pdf

Left-Side Equation Numbers

Left-numbered equation:
 (1)
$$e^{i\pi} + 1 = 0$$

Another equation:
 (2)
$$F = G\frac{m_1 m_2}{r^2}$$

The screenshot shows a LaTeX editor window with the following code:

```

lab03 > %* Exo3_8_6.tex > ...
1 \documentclass[fleqn,leqno]{article}
2 \usepackage[T1]{fontenc}
3 \usepackage{amsmath}
4
5 \begin{document}
6
7 \section*{Left-Aligned Equations with Left-Side Numbers}
8
9 \begin{equation}
10 \frac{\partial^2 \psi}{\partial t^2} = c^2 \frac{\partial^2 \psi}{\partial x^2}
11 \end{equation}
12
13 \begin{equation}
14 \nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}
15 \end{equation}
16
17 \end{document}
18
19

```

The rendered output shows the title "Left-Aligned Equations with Left-Side Numbers" followed by two equations, each preceded by a left-aligned number in parentheses:

- (1) $\frac{\partial^2 \psi}{\partial t^2} = c^2 \frac{\partial^2 \psi}{\partial x^2}$
- (2) $\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}$

5. Расширенное использование amsmath / Using Mathtools

The screenshot shows a LaTeX editor window with the following code:

```

lab03 > %* Exo3_8_7.tex > ...
1 \documentclass{article}
2 \usepackage[T1]{fontenc}
3 \usepackage{amsmath}
4
5 \begin{document}
6
7 \section*{Advanced amsmath Environments}
8
9 Multiple alignment:
10 \begin{align*}
11 x &= y + z & a &= b + c \\
12 x^2 &= y^2 + 2yz + z^2 & a^2 &= b^2 + 2bc + c^2
13 \end{align*}
14
15 Equation gathering:
16 \begin{gather*}
17 e^{i\theta} = \cos\theta + i\sin\theta \\
18 \cos^2\theta + \sin^2\theta = 1 \\
19 \tan\theta = \frac{\sin\theta}{\cos\theta}
20 \end{gather*}
21
22 Multiline equation:
23 \begin{multline*}
24 p(x) = 3x^6 + 14x^5y + 590x^4y^2 + 19x^3y^3 \\
25 - 12x^2y^4 - 12xy^5 + 2y^6 - a^3b^3
26 \end{multline*}
27
28 \end{document}

```

The rendered output shows the title "Advanced amsmath Environments" followed by three sections:

- Multiple alignment:**

$$\begin{array}{rcl} x & = & y + z \\ x^2 & = & y^2 + 2yz + z^2 \end{array} \qquad \begin{array}{rcl} a & = & b + c \\ a^2 & = & b^2 + 2bc + c^2 \end{array}$$
- Equation gathering:**

$$\begin{aligned} e^{i\theta} &= \cos\theta + i\sin\theta \\ \cos^2\theta + \sin^2\theta &= 1 \\ \tan\theta &= \frac{\sin\theta}{\cos\theta} \end{aligned}$$
- Multiline equation:**

$$p(x) = 3x^6 + 14x^5y + 590x^4y^2 + 19x^3y^3 - 12x^2y^4 - 12xy^5 + 2y^6 - a^3b^3$$

6. Математика выделена жирным шрифтом с bm / Math in bold with bm

The screenshot shows a LaTeX editor window with a source code editor on the left and a preview window on the right. The source code defines a section titled "Bold Mathematics" and demonstrates the use of the `\boldmath` and `\bm` commands to format mathematical symbols in bold. The preview window shows the rendered output, including the section title and the formatted equations.

Source Code:

```

8 \section*{Bold Mathematics}
9
10 With \verb|\boldmath|:
11 {\boldmath
12 \[
13 \nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}
14 \]}
15 }
16
17 With \verb|\bm|:
18 \[
19 \bm{\nabla} \times \bm{E} = -\frac{\partial \bm{B}}{\partial t}
20 \]
21
22 Mixed bold and normal:
23 \[
24 \bm{F} = q(\bm{E} + \bm{v} \times \bm{B})
25 \]
26
27 \end{document}

```

Preview Output:

Bold Mathematics

With `\boldmath`:

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

With `\bm`:

$$\bm{\nabla} \times \bm{E} = -\frac{\partial \bm{B}}{\partial t}$$

Mixed bold and normal:

$$\bm{F} = q(\bm{E} + \bm{v} \times \bm{B})$$

Выводы

В ходе лабораторной работы №3 я изучил основы набора математических выражений в LaTeX, познакомился с пакетами `amsmath`, `mathtools`, `bm`, и `unicode-math`. В результате я научился выравнивать уравнения, изменять математические шрифты, делать символы жирными и работать с многострочными выражениями.

As a result, the goal of the lab was achieved: mastering math mode in LaTeX and using key math packages for professional-quality typesetting.

Список литературы

Спасибо за внимание
